

AN *AB INITIO* ROVIBRATIONAL LINELIST FOR HPO

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HPO appears important as a key intermediate species for phosphine synthesis on Venus (Mráziková et al., *Astrobiology*, **24**(4), 407 (2024)) and is included in the modeling of 55 Cancri e (Zilinskas et al., *A&A*, **697**, A34 (2025)). Here we present the production of an *ab initio* rotation-vibration line list for the HPO molecule in its ground electronic state. The potential energy and dipole moment surfaces are computed using CCSD(T)-F12b with the cc-pVQZ-F12 and aug-cc-pV(5+d)Z basis sets, respectively, on a grid of 5000 geometries spanning energies up to 20000 cm⁻¹. Variational calculations performed using the nuclear motion program TROVE include total angular momentum states up to $J = 150$ with energies spanning up to 30,000 cm⁻¹ and transitions covering the wavenumber range up to 20,000 cm⁻¹. Fundamental vibrational frequencies are in good agreement with limited experimental and earlier theoretical values. This is the first comprehensive line list for HPO applicable for a large range of temperatures. It will be included in the ExoMol database.